

Perdiction Results

2024-10-17

Bill Passage

- **Number of observations:** 40,000.

Table 1: True Bill Passage Rates

Chamber	True Passage Rate
Pass House	0.074
Pass Senate	0.060

Table 2: LLM Bill Passage Rates

Chamber	Model	Prompt	Passage Rate
Pass House	GPT-3.5	w/o date restriction	0.899
Pass House	GPT-3.5	w/ date restriction	0.885
Pass House	GPT-4o	w/o date restriction	0.043
Pass House	GPT-4o	w/ date restriction	0.384
Pass Senate	GPT-3.5	w/o date restriction	0.801
Pass Senate	GPT-3.5	w/ date restriction	0.783
Pass Senate	GPT-4o	w/o date restriction	0.042
Pass Senate	GPT-4o	w/ date restriction	0.330

Table 3: Performance Evaluation of LLM Predictions of Bill Passage

PredictedOutcome	Model	Prompt	Accuracy	TPR	FPR
Pass House	GPT-3.5	w/o date restriction	0.173	0.985	0.892
Pass House	GPT-3.5	w/ date restriction	0.186	0.984	0.877
Pass House	GPT-4o	w/o date restriction	0.912	0.198	0.031
Pass House	GPT-4o	w/ date restriction	0.644	0.691	0.359
Pass Senate	GPT-3.5	w/o date restriction	0.256	0.973	0.790
Pass Senate	GPT-3.5	w/ date restriction	0.274	0.975	0.771
Pass Senate	GPT-4o	w/o date restriction	0.925	0.225	0.031
Pass Senate	GPT-4o	w/ date restriction	0.695	0.711	0.306

Figure 1: Accuracy of Bill Passage Predictions, GPT-4o

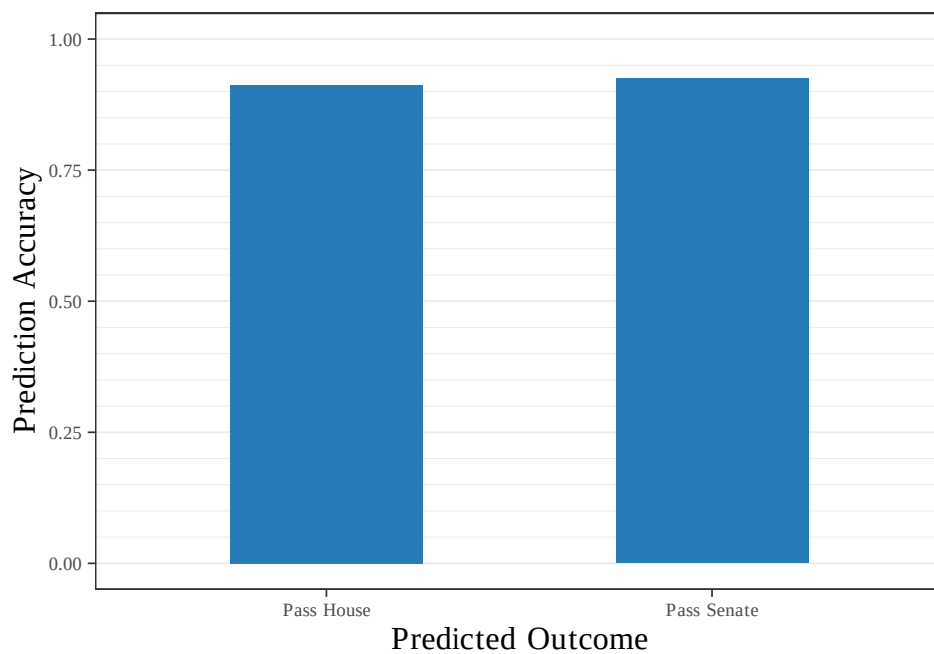


Figure 2: Accuracy of Bill Passage Predictions, GPT-4o, by Prompt

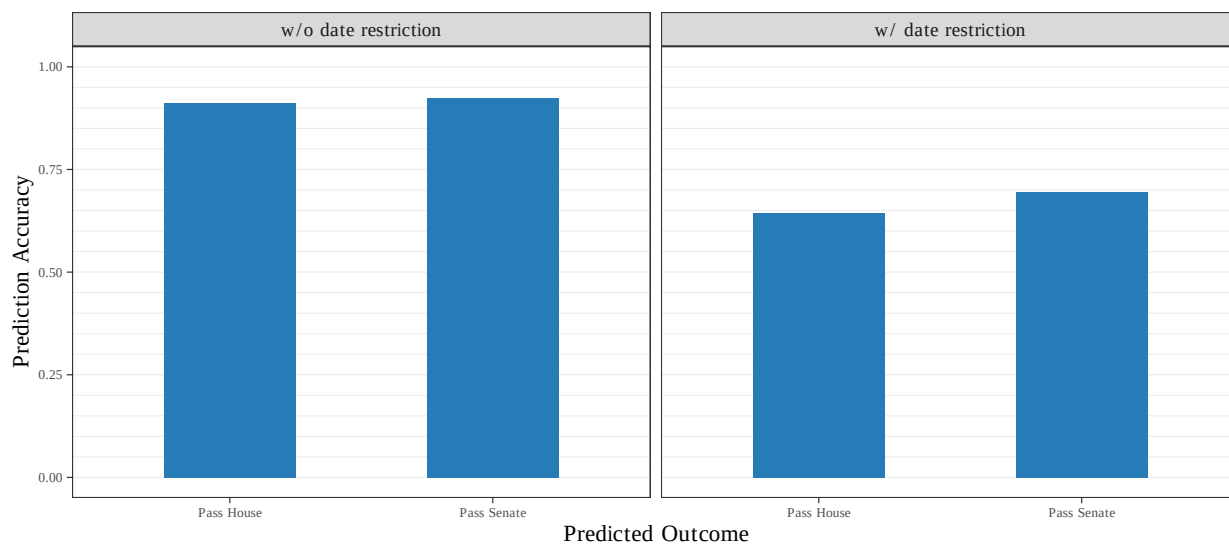


Figure 3: TPR of Bill Passage Predictions, GPT-4o, by Prompt

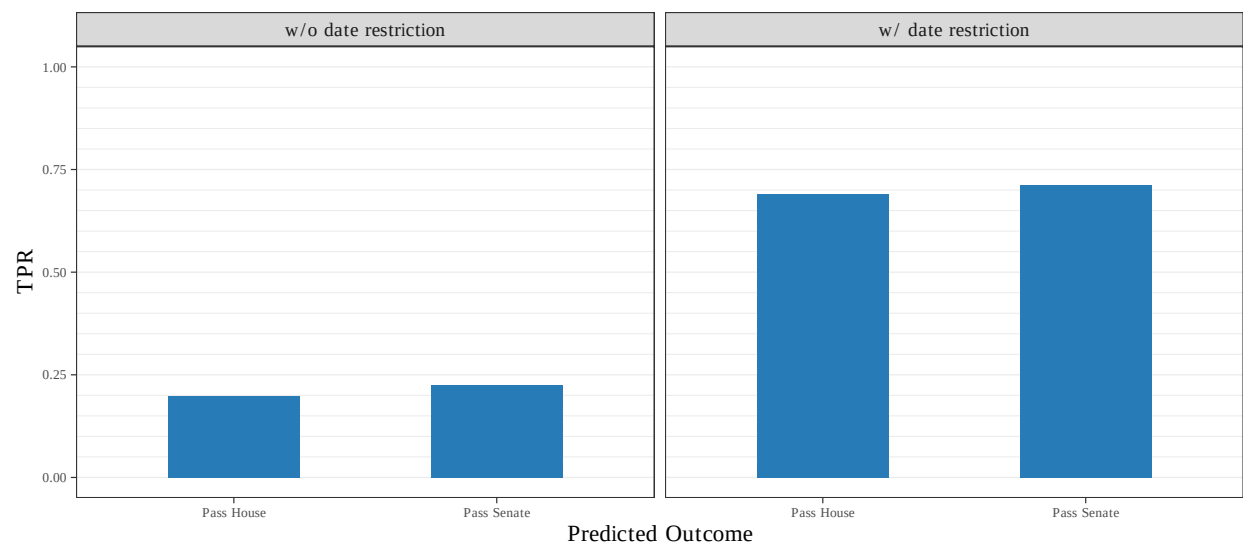
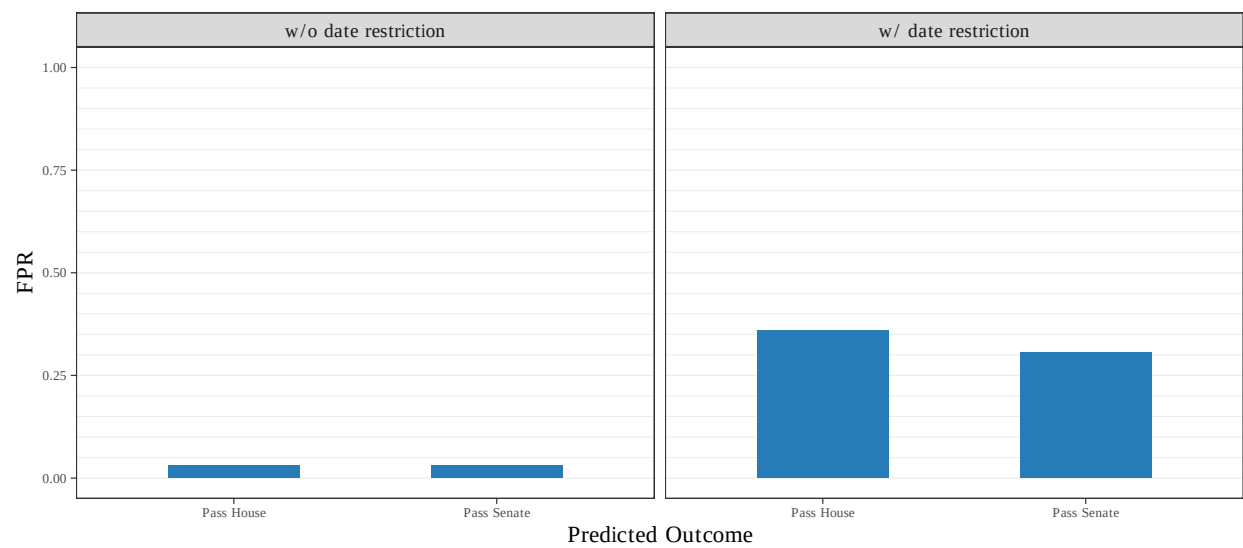


Figure 4: FPR of Bill Passage Predictions, GPT-4o, by Prompt



Bill Summary Completion

- **Number of Observations:** 40,000, with 39 responses exceeding the 750-token limit.
- **Cleaning:** For responses missing the provided part of the bill summary, we append it. If a response starts differently from the provided summary, we leave it as is. Both the true and LLM summaries are cleaned by removing **non-alphanumeric** characters and converting everything to lowercase.
- **Embedding Model:** Cleaned summaries are embedded using OpenAI’s `text-embedding-3-small` model, with an embedding size of 1536.

Examples of Successful Completion

Here are 10 examples where GPT-4o is considered to successfully complete the bill summary. Specifically, the cleaned responses are identical to the cleaned true summaries.

Table 4: Examples of Successful Completion of Bill Summaries (Non-Embedded)

Model	Prompt	DescriptionClean	DescriptionLLMClean
GPT-4o	w/o date restriction	a bill to improve and reauthorize provisions relating to the application of the antitrust laws to the award of needbased educational aid	a bill to improve and reauthorize provisions relating to the application of the antitrust laws to the award of needbased educational aid
GPT-4o	w/o date restriction	to amend title xviii of the social security act to distribute additional information to medicare beneficiaries to prevent health care fraud and for other purposes	to amend title xviii of the social security act to distribute additional information to medicare beneficiaries to prevent health care fraud and for other purposes
GPT-4o	w/o date restriction	a bill to amend the comprehensive environmental response compensation and liability act of 1980 to promote the cleanup and reuse of brownfields to provide financial assistance for brownfields revitalization to enhance state response programs and for other purposes	a bill to amend the comprehensive environmental response compensation and liability act of 1980 to promote the cleanup and reuse of brownfields to provide financial assistance for brownfields revitalization to enhance state response programs and for other purposes
GPT-4o	w/o date restriction	a bill to provide a national medal for public safety officers who act with extraordinary valor above and beyond the call of duty and for other purposes	a bill to provide a national medal for public safety officers who act with extraordinary valor above and beyond the call of duty and for other purposes
GPT-4o	w/o date restriction	a bill to reauthorize subtitle a of the victims of child abuse act of 1990	a bill to reauthorize subtitle a of the victims of child abuse act of 1990
GPT-4o	w/ date restriction	a bill to authorize the federal programs to prevent violence against women and for other purposes	a bill to authorize the federal programs to prevent violence against women and for other purposes
GPT-4o	w/ date restriction	to designate the federal building located at 200 federal plaza in paterson new jersey as the robert a roe federal building	to designate the federal building located at 200 federal plaza in paterson new jersey as the robert a roe federal building
GPT-4o	w/ date restriction	a bill to repeal the public utility holding company act of 1935 to enact the public utility holding company act of 1997 and for other purposes	a bill to repeal the public utility holding company act of 1935 to enact the public utility holding company act of 1997 and for other purposes
GPT-4o	w/ date restriction	a bill to provide antiretaliation protections for antitrust whistleblowers	a bill to provide antiretaliation protections for antitrust whistleblowers
GPT-4o	w/ date restriction	a bill to designate the federal building located at 709 west 9th street in juneau alaska as the hurff a saunders federal building	a bill to designate the federal building located at 709 west 9th street in juneau alaska as the hurff a saunders federal building

Tables and Figures

Table 5: Performance Evaluation of LLM Completion of Bill Summaries (Non-Embedded)

Model	Prompt	Average
GPT-3.5	w/o date restriction	0.010
GPT-3.5	w/ date restriction	0.015
GPT-4o	w/o date restriction	0.034
GPT-4o	w/ date restriction	0.033

Table 6: Performance Evaluation of LLM Completion of Bill Summaries (Embedded)

Metric	Model	Prompt	Average	Baseline	Relative to Baseline
Cosine Similarity	GPT-3.5	w/o date restriction	0.810	0.379	1.134
Cosine Similarity	GPT-3.5	w/ date restriction	0.814	0.379	1.144
Cosine Similarity	GPT-4o	w/o date restriction	0.830	0.379	1.188
Cosine Similarity	GPT-4o	w/ date restriction	0.829	0.379	1.185
Euclidean Distance	GPT-3.5	w/o date restriction	0.580	1.110	0.478
Euclidean Distance	GPT-3.5	w/ date restriction	0.573	1.110	0.484
Euclidean Distance	GPT-4o	w/o date restriction	0.536	1.110	0.517
Euclidean Distance	GPT-4o	w/ date restriction	0.538	1.110	0.515

Figure 5: Cosine Similarity Between True and LLM Bill Summaries Compared to a Random Baseline, GPT-4o, by Prompt

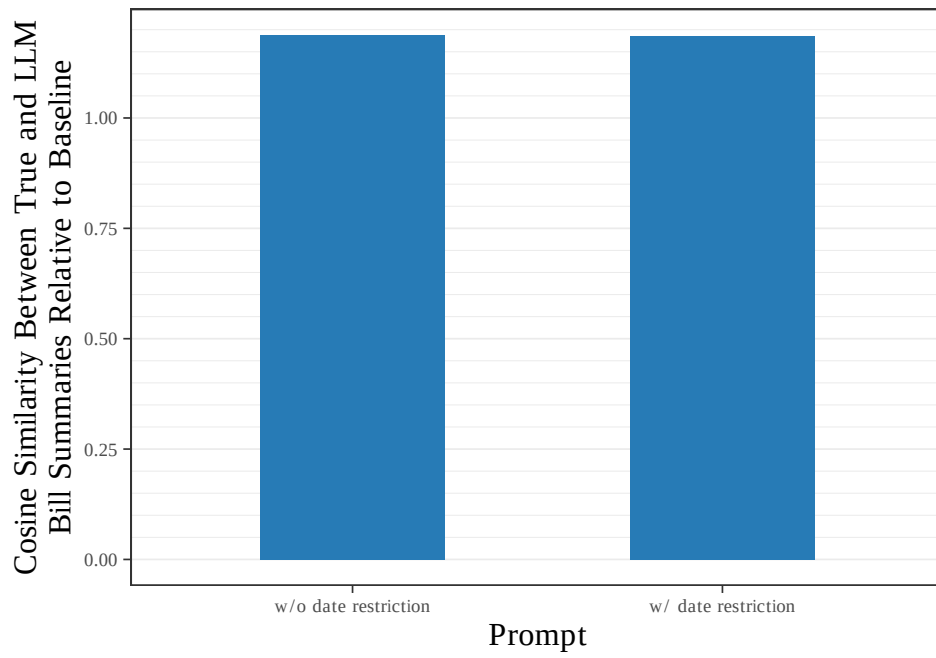


Figure 6: Euclidean Distance Between True and LLM Bill Summaries Compared to a Random Baseline, GPT-4o, by Prompt

